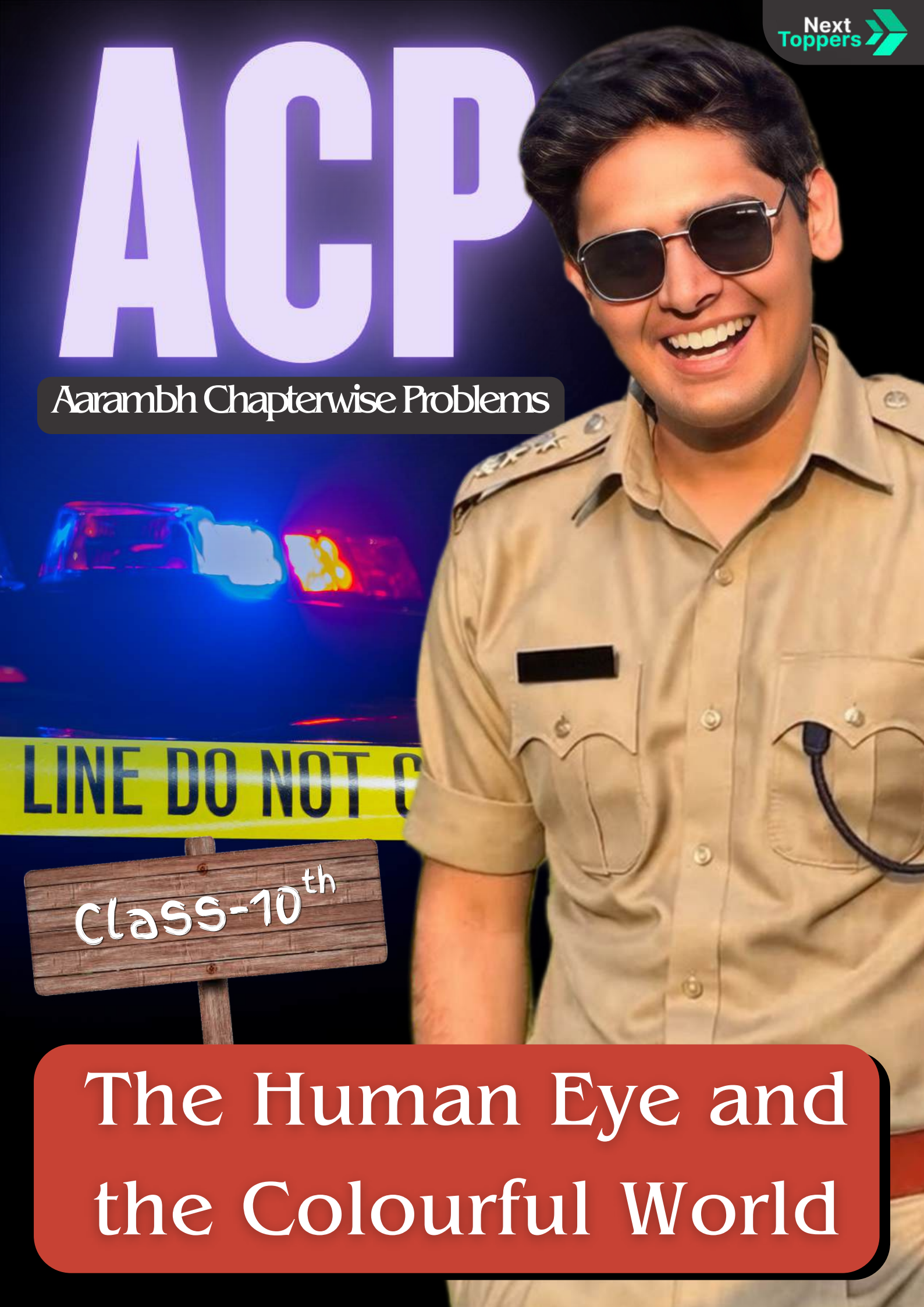


ACP

Aarambh Chapterwise Problems



The Human Eye and
the Colourful World

The Human Eye and the Colourful World

PART 1 (Basic MCQs)

Read the questions carefully!

1. When light rays enter the eye, most of the refraction occurs at the
 - (a) Crystalline lens
 - (b) The outer surface of the cornea
 - (c) Iris
 - (d) Pupil
2. In which part of the human eye is the image of an object formed?
 - (a) Iris
 - (b) Pupil
 - (c) Retina
 - (d) Cornea
3. The danger signals installed at the top of tall buildings are red in colour. These can be easily seen from a distance because, among all other colours, the red light
 - (a) is scattered the most by smoke or fog
 - (b) is scattered the least by smoke or fog
 - (c) is absorbed the most by smoke or fog
 - (d) moves fastest in the air
4. Which of the following phenomena of light are involved in the formation of a rainbow?
 - (a) Reflection, refraction and dispersion
 - (b) Refraction, dispersion and total internal reflection
 - (c) Refraction, dispersion and internal reflection
 - (d) Dispersion, scattering and total internal reflection
5. A person sees an object closer to his eyes. What changes will take place in his eyes?
 - (a) the pupil size will expand
 - (b) the ciliary muscles will contract
 - (c) the focal length of the eye lens will increase
 - (d) the amount of light entering the eye will increase
6. The splitting of white light into different colours on passing through a prism is called
 - (a) reflection
 - (b) refraction
 - (c) dispersion
 - (d) deviation
7. Which of the following phenomena contributes significantly to the reddish appearance of the sun at sunrise or sunset?
 - (a) Dispersion of light

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- (b) Scattering of light
- (c) Total internal reflection of light
- (d) Reflection of light from the earth

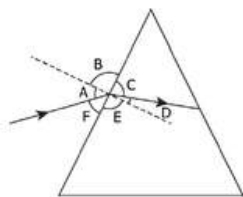
8. Why do stars appear to twinkle at night?

- (a) Because the light of stars travels in a different medium
- (b) Because the distance of a star varies when the earth rotates
- (c) Because the star changes its position relative to the earth
- (d) Because of atmospheric refraction by layers of air of different densities

9. The image shows a light ray incident on a glass prism.

The various angles are labelled in the image. Which angle shows the angle of incidence and angle of refraction, respectively?

- (a) A and D
- (b) B and E
- (c) C and F
- (d) D and F



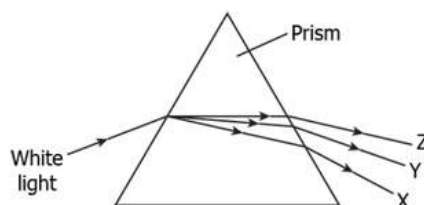
10. A person gets out in the sunlight from a dark room. How does his pupil regulate and control the light entering the eye?

- (a) The size of the pupil will decrease, and less light will enter the eye
- (b) The size of the pupil will decrease, and more light will enter the eye
- (c) The size of the pupil will remain the same, but more light will enter the eye
- (d) The size of the pupil will remain the same, but less light will enter the eye

11. The image shows the dispersion of the white light in the prism.

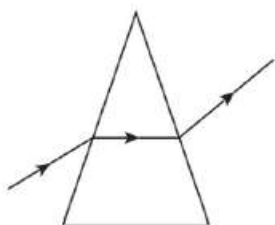
What will be the colours of the X, Y and Z?

- (a) X: red; Y: green; Z: violet
- (b) X: violet; Y: green; Z: red
- (c) X: green; Y: violet; Z: red
- (d) X: red; Y: violet; Z: green

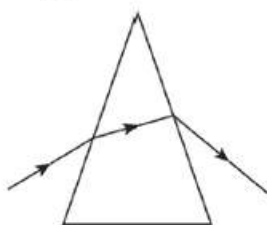


12. Which image shows the deviation of light in a prism?

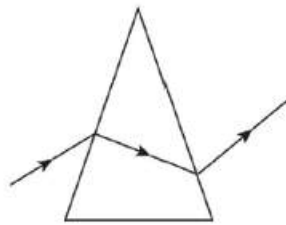
(a)



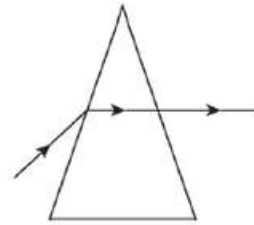
(b)



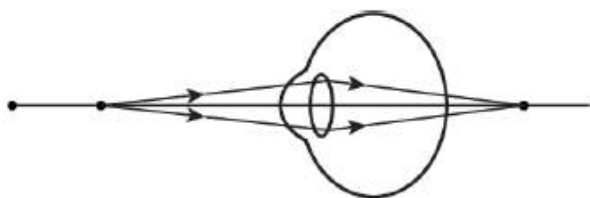
(c)



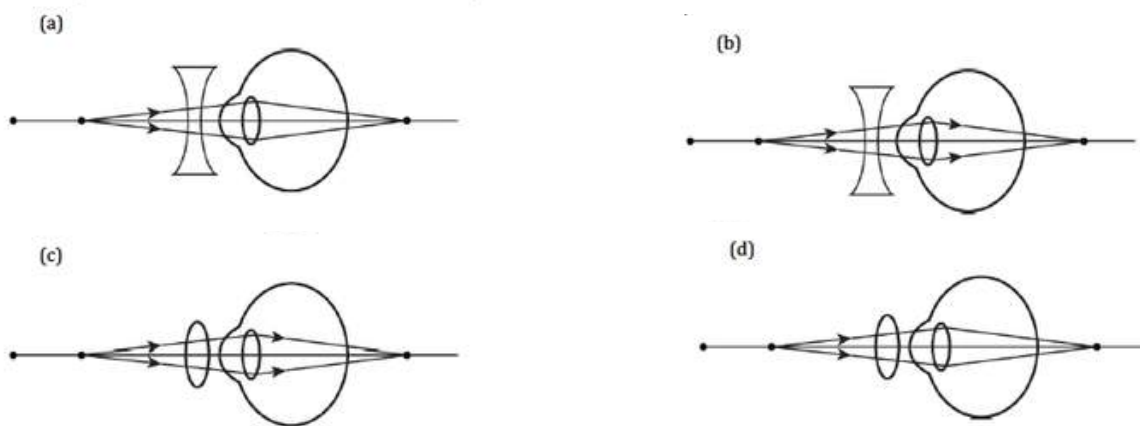
(d)



13. The image shows the ray diagram of a defective eye.

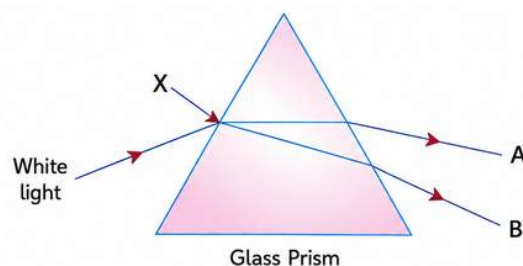


Which option shows the correction of the defect of the eye?



14. A glass prism is shown. White light is incident on the prism and emerges as two rays A and B. Choose the correct option for the colour of rays A and B.

Option	Colour of Ray A	Colour of Ray B
(a)	Blue	Red
(b)	Green	Yellow
(c)	Red	Violet
(d)	Violet	Indigo

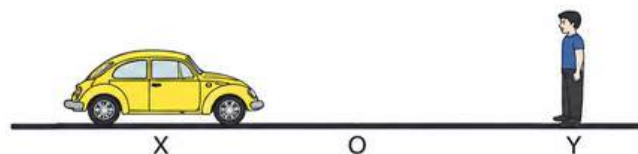


15. A person is reading a book under dim light for a long time. Which of the following changes occur in the eye to help him see the book clearly?

- (a) Pupil contracts and ciliary muscles relax.
- (b) Pupil dilates and ciliary muscles contract.
- (c) Pupil contracts and eye lens becomes thinner.
- (d) Pupil remains unchanged and eye lens becomes thinner.

16. A person standing at point Y is watching a car coming from a point X to O as shown. The variation in the parts of eye while seeing the car at X and O.

- (I) At X the focal length is higher than at O.
- (II) At O the focal length is higher than at X.
- (III) At X, the eye lens is thicker than at O.
- (IV) At O the eye lens is thicker than at X.



Which changes in the person's eye would likely occur while watching the car?

- (a) (I) and (III)
- (b) (I) and (IV)
- (c) (II) and (III)
- (d) (II) and (IV)

17. The face of the Moon that is visible to us is called the near side and the face of the Moon which is invisible to us is called the far side.

What colour would the sky appear to an astronaut standing on the 'far side' of the Moon and why?

- (a) Blue, as the Moon's atmosphere scatters sunlight just like Earth.
- (b) White, as the Moon's surface reflects all the light that falls on it.
- (c) Black, as there is no atmosphere on the Moon to scatter sunlight.
- (d) Black, as sunlight does not fall on the far side of the Moon.

**ASSERTION AND REASON TYPE QUESTION (1 Marks)**

In the following questions a statement of Assertion is followed by a statement of Reason.

Mark the correct choice as two statements are given one labeled Assertion (A) and the other labeled Reason (R).

Select the correct answer to these questions from the codes (a), (b), (c) and (d) as given below:

- (a) Both A and R are true, and R is correct explanation of the assertion.
- (b) Both A and R are true, but R is not the correct explanation of the assertion.
- (c) A is true, but R is false.
- (d) A is false, but R is true.

18 Assertion (A): Increase in the curvature of the eye lens can cause distant objects to be focused before the retina.

Reason (R): An increase in the curvature of the eye lens decreases its focal length, causing distant objects to be focused in front of the retina.

19. Assertion (A): A person wearing spectacles for hypermetropia may not need them while viewing distant objects.

Reason (R): Hypermetropia mainly affects the ability of the eye to focus on nearby objects.

20. Assertion (A): During accommodation for viewing a nearby object, the power of the eye lens increases.

Reason (R): Contraction of ciliary muscles increases the curvature of the eye lens, thereby reducing its focal length.

21. Assertion (A): If the Earth's atmosphere did not exist, stars would not appear to twinkle.

Reason (R): Twinkling is caused by continuous atmospheric refraction due to layers of air having different optical densities.

22. Assertion: The Sun appears reddish at sunrise and sunset.

Reason: At sunrise and sunset sunlight travels a longer path through the atmosphere and shorter wavelengths are scattered away.

23. Assertion (A): A prism can produce a spectrum even when monochromatic light is incident on it.

Reason (R): Dispersion occurs because different wavelengths are refracted through different angles.

24. Assertion (A): Red light is preferred for danger signals because it is scattered the least by atmospheric particles.

Reason (R): Red light has the longest wavelength and is scattered the least by atmospheric particles.

25. Assertion (A): Due to scattering of sunlight, the clear sky predominantly appears blue.

Reason (R): The wavelength of blue light is nearly 1.8 times the wavelength of red light.

26. Assertion (A): When objects are observed through hot air, they appear to be moving slightly.

Reason (R): Cool air is optically denser; hot air is optically rarer.

27. Assertion (A): A normal human eye cannot see objects clearly when they are placed closer than about 25 cm.

Reason (R): For objects closer than 25 cm, the ciliary muscles cannot contract further to decrease the focal length of the eye lens.

PART 2 (Subjective Questions)

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SHORT ANSWER TYPE QUESTIONS (2 and 3 Marks)

28. (a) What is visible spectrum?
 (b) Why is red used as the stopping light at traffic signals?
 (c) Two identical glass prisms are placed inverted with respect to each other.
 A light beam is passed through one side of the combination. Will there be any dispersion? Justify your answer.

OR

Due to the gradual weakening of the ciliary muscles and loss of flexibility of the eye lens, a person develops a defect of vision.

- (a) Name the defect.
 (b) State its cause.
 (c) How is this defect corrected?
 (d) If the person also suffers from myopia, name the lens used and explain its structure.

29. (i) A person is suffering from both myopia and hypermetropia.
 (a) What kind of lenses can correct this defect?
 (b) How are these lenses prepared?
 (ii) A person needs a lens of power $+3D$ for correcting his near vision and $-3D$ for correcting his distant vision. Calculate the focal lengths of the lenses required to correct these defects.

30. What is scattering of light? Why is the colour of the clear sky blue? Explain.

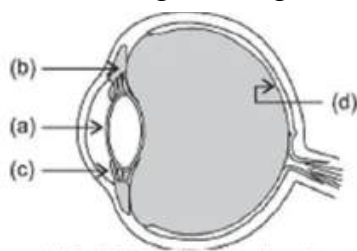
OR

Tiny droplets of water are responsible for the natural spectrum formed in the sky after a rain shower. What is that spectrum known as? Draw a labelled diagram to show the formation of a rainbow.

31. Explain the process of accommodation of the human eye. What changes occur in the eye while viewing:
 (a) a nearby object
 (b) a distant object?

32. Explain why:
 (a) the sky appears blue during the day.
 (b) the Sun appears reddish at sunrise and sunset.

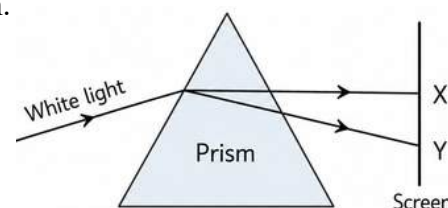
33. Name the four parts labelled as a, b, c and d in the given diagram and write their functions.



34. What is myopia (near-sightedness)? Draw a ray diagram to show how it can be corrected using a lens.
 35. What is hypermetropia (far-sightedness)? Draw a ray diagram to show how this defect can be corrected using a lens.

36. In the figure given below, a narrow beam of white light is shown to pass through a triangular glass prism. After passing through the prism, it produces a spectrum XY on the screen.

- Name the phenomenon.
- State the colours seen at X and Y.
- Why do different colours of white light bend at different angles through a prism?



37. Differentiate between a glass slab and a glass prism. What happens when a narrow beam of

- monochromatic light, and
 - white light
- passes through (a) a glass slab and (b) a glass prism?

38. The near point of the eye of a person is 50 cm. Find the power and the corrective lens required by the person to enable him to see clearly the objects placed at 25 cm from the eye.

LONG ANSWER TYPE QUESTIONS (5 Marks)



(pahle points socho fir likho)

39. The students in a class took a thick sheet of cardboard and made a small hole in its centre. Sunlight was allowed to fall on this small hole and they obtained a narrow beam of white light. A glass prism was taken and this white light was allowed to fall on one of its faces. The prism was turned slowly until the light that comes out of the opposite face of the prism appeared on the nearby screen. They studied this beautiful band of light and concluded that it is a spectrum of white light.

- Give any one more instance in which this type of spectrum is observed.
- What happens to white light in the above case?
- List two conditions necessary to observe a rainbow.

40. A person uses lenses of power -0.5 D in his spectacles for the correction of his vision.

- Name the defect of vision the person is suffering from.
- List two causes of this defect.
- Determine the focal length of the lenses used in the spectacles.

41. What is atmospheric refraction? Use this phenomenon to explain the following natural events.

- Twinkling of stars
- Advance sunrise and delayed sunset.. Draw diagrams to illustrate your answers

42. (a) What is presbyopia? State its cause. How is it corrected?

(b) Why does the sun appear reddish early in the morning? Explain with the help of a labelled diagram.

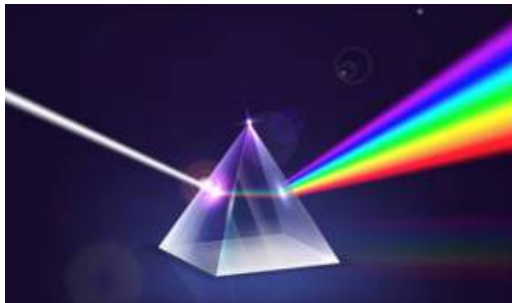
43. State the cause of dispersion of white light by a glass prism. How did Newton, using two identical glass prisms, show that white light is made of seven colours? Draw a ray diagram to show the path of a narrow beam of white light, through a combination of two identical prisms arranged together in an inverted position with respect to each other, when it is allowed to fall obliquely on one of the faces of the first prism of the combination.

44. (a) A student is unable to see clearly the words written on the blackboard placed at a distance of approximately 3 m from him. Name the defect of vision the boy is suffering from. State the possible causes of this defect and explain the method of correcting it. (b) Why do stars twinkle? Explain.

PART 3 (Advanced Questions)

Case Study/Source Based Question (5 Marks)

45. Aarush is a keen observer and loved the spectacular colours in a rainbow. He also observed the same pattern when he allowed sunlight to pass through a glass prism. He guessed that it is due to the inclined refracting surfaces of a glass prism which is responsible for showing such exciting phenomenon. Whereas, no such phenomenon was observed when light passes through a glass slab.



- (a) Name the optical phenomena involved in rainbow formation.
- (b) What is the band of colours known as? Explain the cause of formation of band of colours?
- (c) Why don't we observe similar band of colours when light passes through a glass slab?

46. We know that when light goes from one medium to another medium having different optical densities, then refraction of light rays (or bending of light rays) takes place. Now, in the atmosphere, we have air everywhere. But all the air in the atmosphere is not at the same temperature. Some of the air layers of the atmosphere are cold whereas other air layers of the atmosphere are comparatively warm (or hotter). Now the cooler air layers of the atmosphere behave as optically denser medium for light rays whereas the warmer air layers (or hotter air layers) of the atmosphere behave as optically rarer medium for the light rays. So, in the same atmosphere we have air layers having different optical densities. And when light rays pass through the atmosphere having air layers of different optical densities, then refraction of light takes place. The refraction of light caused by the Earth's atmosphere (having air layers of varying optical densities) is called atmospheric refraction.

i. With respect to atmospheric refraction which of the following point distinguish between cold air and hot air?

- (a) Cold air is denser than hot air
- (b) Hot air is lighter than cold air
- (c) Cold air has higher refractive index than hot air.
- (d) All of the above

ii. What is the reason behind twinkling of Stars?

- (a) Dispersion of star light
- (b) Reflection of star light
- (c) Refraction of star light
- (d) All of the above

iii. Why Sun appears flattened during sunrise and sunset?

- (a) Because Sun is closer to Earth
- (b) Because Earth is rotating
- (c) Because Earth is revolving
- (d) Because of atmospheric refraction

iv. How much time from sunrise to sunset is lengthened because of atmospheric refraction?

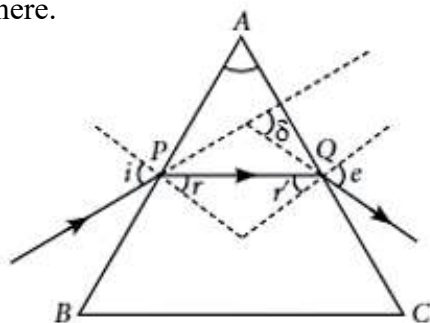
- (a) 4 hours
- (b) 2 minutes
- (c) 4 minutes
- (d) 2 hours

Or

iv. When light rays from Stars enter into Earth's atmosphere, it travels from:

- (a) denser to rarer medium
- (b) rarer to denser medium
- (c) rarer medium to vacuum
- (d) denser medium to vacuum

47. A prism is a transparent refracting medium bounded by two plane surfaces inclined to each other at a certain angle. The refraction of light through a prism follows the laws of refraction. In the prism, refraction takes place on its refracting surface, meaning when the light enters the prism and when the light leaves the prism. The refraction through a prism is shown here.



Here, $\angle A$ is the angle of the prism, $\angle i$ is the angle of incidence at face AB, and $\angle e$ is the angle of emergence at the other face AC. The incident ray suffers a deviation or bending through an angle δ due to the refraction through the prism. This angle is called the angle of deviation, as shown in the figure.

$$\angle i + \angle e = \angle \delta + \angle A$$

i. The angle between the two refracting surfaces of a prism is called

- (a) Angle of prism
- (b) Angle of incidence
- (c) Angle of deviation
- (d) Angle of emergence

ii. The angle between the direction of the incident ray produced forward and the emergent ray is called

- (a) Angle of emergence
- (b) Angle of deviation
- (c) Angle of incidence
- (d) None of these

iii. When a ray is refracted through a prism, then

- (a) $\angle i = \angle \delta$
- (b) $\angle i = \angle e + \angle \delta$
- (c) $\angle \delta = \angle e$
- (d) $\angle i > \angle r$

iv. The angle of deviation depends on

- (a) Refractive index of prism
- (b) Angle of incidence
- (c) Both (a) and (b)
- (d) None of these

OR

iv. The two inclined rectangular faces through which light enters and emerges are called

- (a) Reflecting surfaces
- (b) Dispersing surfaces
- (c) Refracting surfaces
- (d) None of these

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48. Sunny and his friends went to the terrace of their apartment building to watch the night sky. They noticed that all apartment buildings had red light installed on the terrace. Sunny had also seen red lights on top of tall towers such as the Pitampura TV Tower and even on top of the GSM towers.

(A) The danger signals installed at the top of tall buildings are red in colour. These can be easily seen from a distance because among all other colours, why?

(B) Sunny wants to know why the Sun appears oval-shaped at sunrise and sunset but appears circular at noon.

(C) Sunny and his friends noticed stars twinkling in the night sky. What causes this phenomenon?



49. We've all seen that part in the movie where the weary desert wanderer has been walking for hours on end, dying of thirst. Then he happens upon a vast body of water on the horizon. He runs towards the water; it grows closer and closer, until he springs himself into the air only to land back down in the sand and no water is in sight. Yet this is due to an optical illusion called Mirage.

(A) Mirage is an example of:

- (a) Reflection of light and refraction of light
- (b) Dispersion of light
- (c) Total internal reflection
- (d) Refraction of light and total internal reflection of light

(B) Mirages are more common in:

- (a) Rainforests
- (b) Dry forests
- (c) Deserts
- (d) Highlands

(C) Mirage is formed because:

- (I) Air above the ground is very hot and air above is cooler.
- (II) Air above the ground is cool and air above is warmer.
- (III) Light rays from a distant object bend towards the normal when coming towards the ground.
- (IV) Light rays from a distant object bend away from the normal when coming towards the ground.

Choose the correct option:

- (a) Both (I) and (III)
- (b) Both (I) and (IV)
- (c) Both (II) and (III)
- (d) Both (II) and (IV)

OR

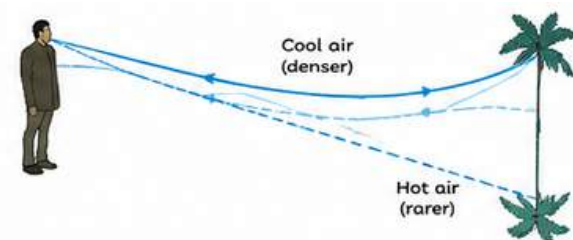
(D) Atmospheric refraction occurs because:

- (a) Refractive index of the medium is gradually changing.
- (b) Presence of dust particles in the atmosphere.
- (c) Large amount of moisture is present in the atmosphere on a humid day.
- (d) Sun's rays travel the most when the Sun is near the horizon.

OR

(D) When sunlight enters the Earth's atmosphere:

- (a) It bends away from the normal.
- (b) It bends towards the normal.
- (c) It first bends towards the normal and then away from the normal.
- (d) It first bends away from the normal and then towards the normal.



- Human Eye – Structure, Functions & Diagram – Q1, Q2, Q10, Q15, Q33
- Accommodation of the Human Eye (Ciliary Muscles, Pupil, Focal Length) – Q5, Q10, Q15, Q16, Q20, Q31
- Myopia (Definition, Causes, Correction, Ray Diagram & Numerical) – Q13, Q18, Q29(i), Q34, Q40, Q44(a)
- Hypermetropia (Definition, Causes, Correction, Ray Diagram & Numerical) – Q19, Q29(i), Q29(ii), Q35, Q38
- Presbyopia (Causes, Correction & Bifocal Lens) – Q28 (OR), Q29(i), Q42(a)
- Defects of Vision (Combined Concepts) – Q27
- Power of Lens & Lens Numerical – Q29(ii), Q38, Q40(c)
- Dispersion of Light, Prism, Spectrum & Newton's Experiment (Including Diagrams) – Q4, Q6, Q9, Q11, Q12, Q14, Q23, Q28(a,c), Q36, Q37, Q39, Q43, Q45, Q47
- Rainbow Formation (Theory & Diagram) – Q4, Q30 (OR), Q39(i,iii), Q45
- Scattering of Light, Blue Sky & Red Colour of Sun – Q3, Q7, Q17, Q22, Q24, Q25, Q28(b), Q30, Q32, Q42(b), Q48(A)
- Atmospheric Refraction, Twinkling of Stars, Advanced Sunrise & Delayed Sunset (Including Diagrams) – Q8, Q21, Q26, Q41, Q44(b), Q46, Q48(B,C), Q49(D)
- Glass Slab vs Glass Prism – Q28(c), Q37, Q45(c)
- Mirage (Atmospheric Refraction & Total Internal Reflection) – Q49



SOLUTIONS

MULTIPLE CHOICE QUESTIONS [MCQ's]

- | | | |
|--------|--------|--------|
| 1.(b) | 11.(b) | 21.(a) |
| 2.(c) | 12.(b) | 22.(a) |
| 3.(b) | 13.(d) | 23.(d) |
| 4.(c) | 14.(c) | 24.(a) |
| 5.(b) | 15.(b) | 25.(c) |
| 6.(c) | 16.(b) | 26.(b) |
| 7.(b) | 17.(c) | 27.(a) |
| 8.(d) | 18.(a) | |
| 9.(a) | 19.(a) | |
| 10.(a) | 20.(a) | |

SHORT ANSWER TYPE QUESTIONS:

28. (a) Visible spectrum is the band of coloured components of a white light beam.
 (b) Red light has the longest wavelength and is scattered the least, so it is visible from a greater distance.
 (c) The combination of two identical prisms behaves like a glass slab, resulting in recombination of the seven colours to produce white light.

OR

- (a) Presbyopia
 b) It occurs due to:
- gradual weakening of the ciliary muscles
 - loss of flexibility of the eye lens
- (c) Presbyopia is corrected using a convex lens for near vision.
 (d) A bifocal lens is used.
- Upper part: Concave lens → for distant vision
 - Lower part: Convex lens → for near vision

29. (i)

(a) Bi-focal lenses can correct both myopia and hypermetropia.

(b) The upper portion of the bi-focal lens consists of a concave lens (to correct myopia) and the lower portion consists of a convex lens (to correct hypermetropia).

(ii)

Given:

- Power for near vision, $P_1 = +3 \text{ D}$
- Power for distant vision, $P_2 = -3 \text{ D}$

Formula:

$$P = 1/f \text{ (where } f \text{ is in metres)}$$

For near vision:

$$f_1 = 1/3 = +0.33 \text{ m} = +33.3 \text{ cm}$$

For distant vision:

$$f_2 = 1/(-3) = -0.33 \text{ m} = -33.3 \text{ cm}$$

Answer:

- Focal length of the lens for near vision = $+0.33 \text{ m}$ (33.3 cm)
- Focal length of the lens for distant vision = -0.33 m (-33.3 cm)

30. Scattering of light: Scattering of light is the phenomenon in which light is redirected in different directions when it strikes particles of a medium.

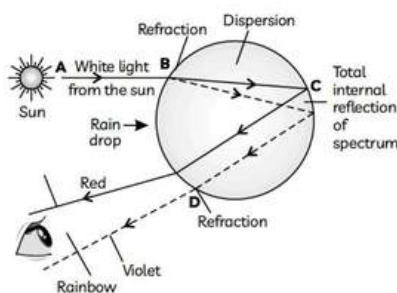
Why is the clear sky blue?

- Sunlight is made up of seven colours (VIBGYOR).
- When sunlight passes through the Earth's atmosphere, tiny air molecules scatter light.
- Blue light has a shorter wavelength and is scattered more than red light.
- The scattered blue light reaches our eyes from all directions, making the clear sky appear blue.

OR

The natural spectrum formed in the sky after a rain shower is called a rainbow.

A rainbow is formed due to refraction, dispersion, internal reflection, and refraction again of sunlight by tiny water droplets present in the atmosphere.



31. Accommodation is the ability of the human eye to adjust the focal length of its eye lens so that objects at different distances are focused clearly on the retina.

(a) While viewing a nearby object:

- The ciliary muscles contract.
- The eye lens becomes thicker (more curved).
- The focal length of the eye lens decreases and its power increases.
- The image is formed clearly on the retina.

(b) While viewing a distant object:

- The ciliary muscles relax.
- The eye lens becomes thinner (less curved).
- The focal length of the eye lens increases and its power decreases.
- The image is formed clearly on the retina.

32. (a) Sky appears blue during the day:

- Sunlight is scattered by tiny molecules and dust particles present in the atmosphere.
- Blue light, having a shorter wavelength, is scattered more than other colours.
- Therefore, the scattered blue light reaches our eyes, making the sky appear blue.

(b) Sun appears reddish at sunrise and sunset:

- At sunrise and sunset, sunlight travels a longer distance through the atmosphere.
- Most of the blue and violet light is scattered away before reaching our eyes.
- Red light, which has the longest wavelength, is scattered the least and reaches our eyes.
- Hence, the Sun appears reddish at sunrise and sunset.

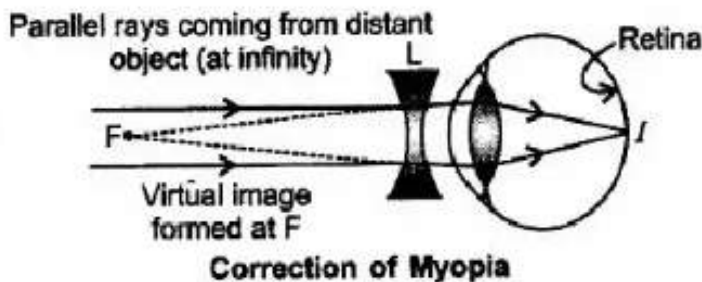
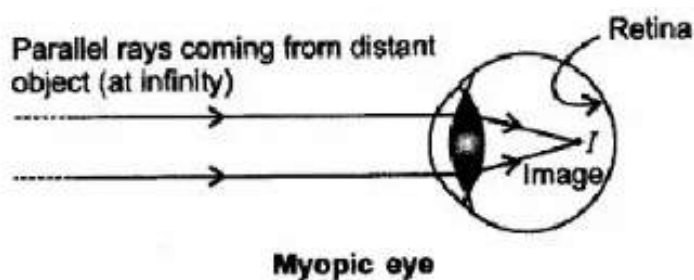
33. (a) Pupil: Regulates the amount of light entering the eye.

(b) Ciliary muscles: Modify the curvature of the eye lens and hence its focal length to focus the image on the retina.

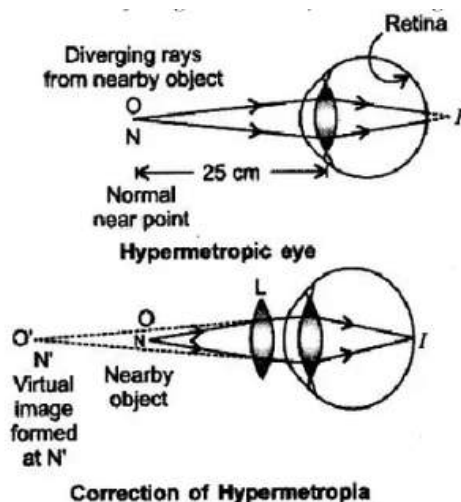
(c) Iris: Controls the size of the pupil.

(d) Retina: A real, inverted and diminished image of the object is formed on it.

34. Myopia is the inability of an eye to view distant objects clearly. In this defect, the image is formed before the retina. For every myopic eye, there exists a far point beyond which clear images cannot be seen. The short-sightedness is corrected by using a concave lens, which diverges the light rays and shifts the image to the retina.



35. Hypermetropia is the inability of an eye to view nearby objects clearly. In this defect, the image is formed beyond the retina. For a hypermetropic eye, there exists a near point farther than the normal near point. It is corrected by using a convex lens, which converges the light rays and shifts the image to the retina.



36. (i) The phenomenon is called dispersion.

(ii)

- At X – Red
- At Y – Violet

(iii) Different colours of white light bend at different angles with respect to the incident beam of light due to the difference in the speed of light of different wavelengths in the prism.

37.	Glass Slab	Glass Prism
	It is rectangular in shape.	It has two triangular faces and three rectangular faces.
	The refracting faces are parallel to each other.	The refracting faces are inclined to each other.
	The emergent ray is parallel to the incident ray (only lateral displacement occurs).	The emergent ray is deviated from the incident ray.

(b) Effect of Monochromatic Light

(i) Through a Glass Slab:

- The ray undergoes lateral displacement.
- The incident and emergent rays remain parallel.
- Dispersion occurs at the first surface but is recombined at the second parallel surface; therefore, no net observable dispersion occurs.

(ii) Through a Glass Prism:

- The ray is deviated due to refraction.
- No dispersion takes place because monochromatic light has only one colour.

(c) (i) Through a Glass Slab:

- White light emerges as white light.
- Dispersion occurs at the first surface but is recombined at the second parallel surface; therefore, no net observable dispersion occurs.
- The incident and emergent rays remain parallel.

(ii) Through a Glass Prism:

- White light splits into its seven constituent colours (VIBGYOR).
- This phenomenon is called dispersion of light.
- Different colours deviate through different angles.

38. Answer. Given: Near point of the person's eye = 50 cm

Object distance, $u = -25$ cm

Image distance, $v = -50$ cm

Using the lens formula = $1/f = 1/v - 1/u$

$$1/f = 1/(-50) - 1/(-25)$$

$$= -1/50 + 2/50$$

$$= 1/50$$

$$f = 50 \text{ cm} = 0.5 \text{ m}$$

Power of the lens = $P = 1/f$ (in metres)

$$= 1/0.5 = +2 \text{ D}$$

Therefore:

- Power of the corrective lens = +2 D
- Corrective lens required = Convex (converging) lens.

39. (i) One more instance: A similar spectrum is observed in a rainbow formed in the sky after a rain shower.
- (ii) What happens to white light: In the prism experiment, white light splits into its seven component colours - Violet, Indigo, Blue, Green, Yellow, Orange and Red (VIBGYOR). This process is called dispersion of light.
- (iii) Conditions necessary to observe a rainbow: The Sun should be shining in one part of the sky (behind the observer). Tiny water droplets must be present in the atmosphere in the direction opposite to the Sun, to refract, disperse and internally reflect sunlight to form the rainbow.

40. (a) The person is suffering from myopia (short-sightedness).

(b) Two causes of myopia are:

1. Excessive curvature (high converging power) of the eye lens.
2. Elongation of the eyeball.

(c) Calculation of focal length:

Given:

Power of lens, $P = -0.5 \text{ D}$

Formula:

$$P = 1/f$$

Therefore,

$$f = 1/P = 1/(-0.5) = -2 \text{ m}$$

The focal length of the lens is -2 m .

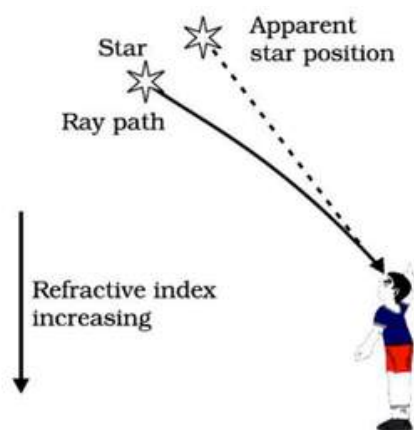
41. Atmospheric Refraction: Atmospheric refraction is the refraction of light by the Earth's atmosphere due to the varying optical densities of different layers of air.

(a) Twinkling of Stars:

Light from a star passes through different layers of the Earth's atmosphere having continuously changing refractive indices. Therefore, starlight undergoes continuous atmospheric refraction.

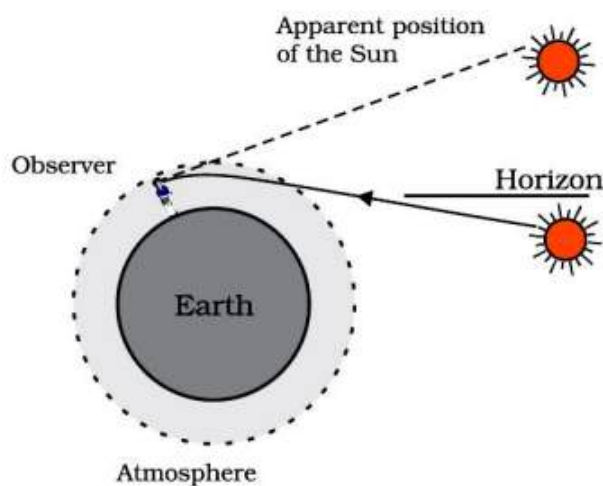
As a result, the apparent position of the star keeps changing slightly from its actual position, and the star generally appears slightly higher than its actual position.

Since the atmospheric conditions keep changing continuously, the apparent position and brightness of the star also keep changing. Sometimes the star appears brighter and sometimes fainter. Hence, stars appear to twinkle.



(b) Advanced Sunrise and delayed sunset -

The sun is visible 2 minutes before sunrise and 2 minutes after sunset because of atmospheric refraction. This can be explained as below. The figure shows the actual position of the sun S at the time of sunrise or sunset, just below the horizon while the apparent position S' , above the horizon as appear to us.



When the sun is slightly below the horizon, the light rays move through the different layers of varying refractive indices of air and get bent towards the normal. These rays appear to come from S' , which is the apparent position of the sun. That is why, the sun is visible to us when it has been actually below the horizon or before the actual crossing of horizon by the sun at the time of sunrise or sunset. So, due to the atmospheric refraction, the phenomenon of advance sunrise and delayed sunset is observed.

42. (a)

(i) Presbyopia:

Presbyopia is a defect of vision that occurs as a part of the normal ageing process. Due to the loss of the power of accommodation of the eye, nearby objects appear blurred. The near point of the eye gradually recedes away. Sometimes, a person may suffer from both myopia and hypermetropia.

(ii) Cause of Presbyopia:

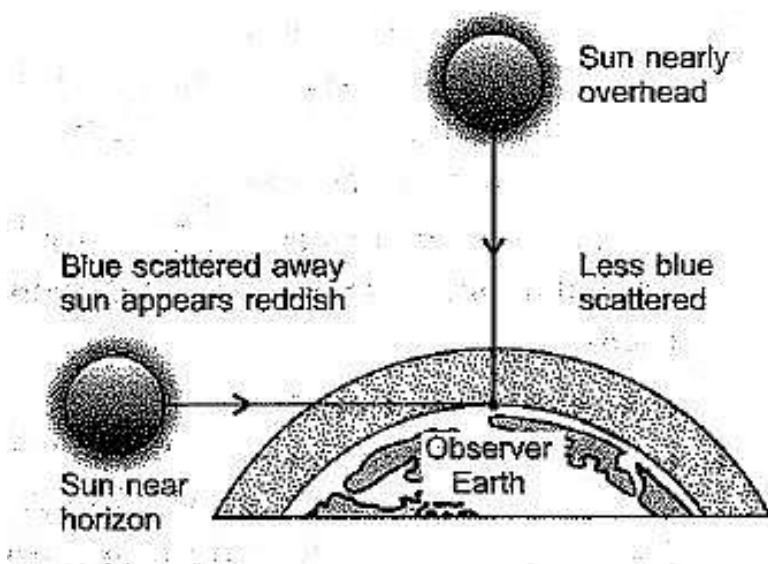
Presbyopia is caused due to the weakening of the ciliary muscles and the loss of elasticity of the eye lens. As a result, the eye lens loses its ability to change its curvature, reducing the power of accommodation.

(iii) Correction of Presbyopia:

Presbyopia is usually corrected by a convex lens for near vision. If a person has both myopia and hypermetropia, then bifocal lenses are used.

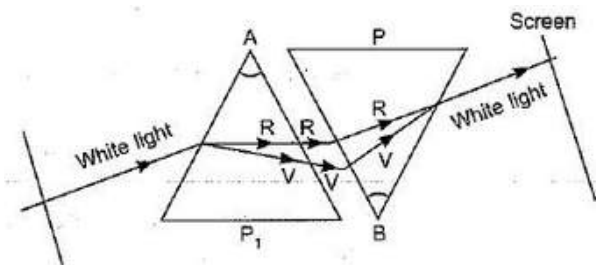
(b) Why does the Sun appear reddish at sunrise and sunset?

At sunrise and sunset, the Sun is near the horizon, so its light has to travel through a greater thickness of the Earth's atmosphere. The tiny particles present in the atmosphere scatter the shorter wavelengths (blue and violet) of sunlight much more than the longer wavelengths. Since red light has the longest wavelength, it is scattered the least and reaches our eyes. Therefore, the Sun appears reddish at sunrise and sunset.



43. Cause of dispersion: From Snell's law of refraction, The refractive index of glass is different for different colours (wavelengths), so different colours bend by different amounts, causing dispersion. Moreover, the refractive index of the material varies inversely with the speed of light and also varies inversely with the wavelength of light. Hence, different colours of white light bend through different angles with respect to the incident light, as they pass through the glass prism.

Newton Experiment: Consider a prism A. When a beam of white light falls obliquely on one of the faces of this prism, it splits up into seven constituent colours. The violet colour deviates the most and the red colour deviates the least. If another identical prism B is placed in an inverted position concerning the first prism A, the constituent coloured rays that emerge out of prism A will be made to merge to come out as a beam of white light, as shown below.



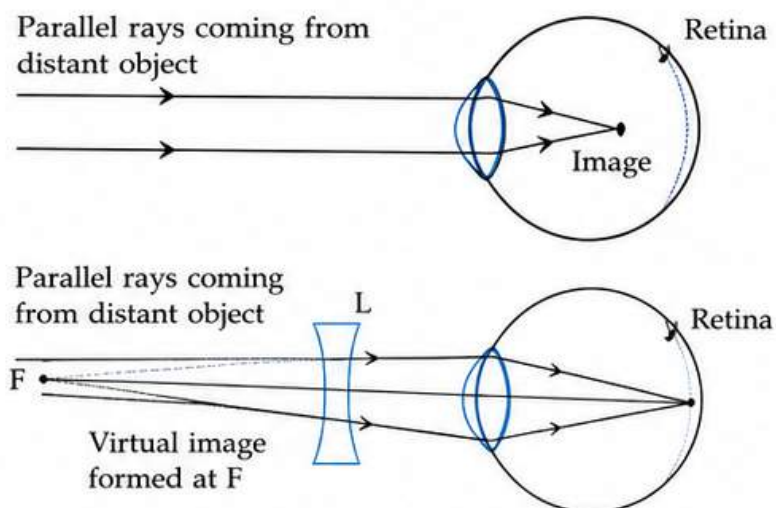
44. (a) The defect of vision is ****Myopia (short-sightedness)****.

Causes:

1. Elongation of the eyeball.
2. Excessive curvature (increased converging power) of the eye lens.

Correction:

Myopia is corrected by using a ****concave (diverging) lens**** of suitable power. The concave lens diverges the incoming light rays so that, after refraction by the eye lens, the image is formed on the retina. This enables the person to see distant objects clearly.



(b) Stars are very far away from the Earth and appear as point sources of light. The light from a star passes through different layers of the Earth's atmosphere, whose refractive index keeps changing due to variations in temperature and density. As a result, the light undergoes atmospheric refraction. This causes the apparent position of the star to change continuously and the amount of light entering the eye to fluctuate. Therefore, stars appear to twinkle.

CASE STUDY/SOURCE BASED QUESTION:

45. (a) Rainbow is formed due to refraction, dispersion and internal reflection of sunlight in water droplets.

(b) The band of colours is called the spectrum. It is formed because different colours of white light are refracted by different amounts when they pass through a glass prism.

(c) We do not observe a similar band of colours when light passes through a glass slab because the refracted colours recombine on emerging from the slab due to its parallel refracting surfaces. Hence, white light is

obtained again.

46. (i) (d)

(ii) (c)

(iii) (d)

(iv) (c)

OR

(iv) (b)

47. (i) (a)

(ii) (b)

(iii) (d)

(iv) (c)

OR

(c)

48.

(A) Red light has the longest wavelength and is scattered the least by air, dust and fog. Therefore, it can travel longer distances and is visible from a greater distance. Hence, danger signals are made red.

(B) At sunrise and sunset, the Sun's rays pass through a thicker layer of the Earth's atmosphere. Due to atmospheric refraction, the lower edge of the Sun is refracted more than the upper edge, making the Sun appear oval or flattened. At noon, the Sun is overhead and its light passes through a shorter path in the atmosphere, so it appears circular.

(C) Stars twinkle due to atmospheric refraction. As the light from stars passes through different layers of the Earth's atmosphere having continuously changing refractive indices, the apparent position and brightness of the stars keep changing. Therefore, stars appear to twinkle.

49.

(A) (d) Refraction of light and total internal reflection of light.

(B) (c) Deserts.

(C) (b) Both (I) and (IV).

OR

(D) (a) Refractive index of the medium is gradually changing.

OR

(D) (b) It bends towards the normal.

